

Augmented Reality In-Field Observation Creation and Visualization in Underperforming Areas

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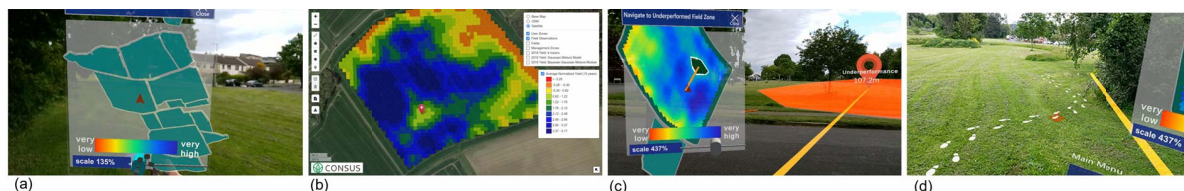


Figure 1: (a) Select The Field for Field Trip From the Minimap; (b) Identify the Underperforming Areas Using the Web-based Interactive Multi-Layer Mapping Tool; (c) Navigate the User to The Underperforming Areas; (d) Record the User's Routine

ABSTRACT

To precisely diagnose and address farming issues that led to crop yield underperformance, agronomists need to conduct field trips at the targeted underperforming areas and record the observed issues to support farming decisions. Traditional field data visualization tools may not provide intuitive visualizations to support field trip investigations at targeted underperforming areas. This paper presents an Augmented Reality in-field observation tool to navigate agronomists to annotate the observed issues at precisely targeted underperforming areas. These recorded observation AR annotations are then synchronized in a Web-based Interactive Multi-Layer Mapping Tool for a complete precision farming decision-making process.

Index Terms: Human-centered computing—Ubiquitous and mobile computing systems and tools—

1 INTRODUCTION

Precision farming (PF) combines information technologies to optimize the crop yield with decreased managerial costs [1]; therefore, various field data collection and visualization approaches have been proposed based on common desktop interfaces, drones, in-field cameras, soil sensors, and Augmented Reality (AR) mobile devices. Compared to other visualization tools, mobile head-mounted AR devices allow for a more instant and intuitive visualization approach to provide expert knowledge needed for farmers anywhere anytime with hands-free interaction capabilities. Therefore, mobile AR data visualization has been applied to increase efficiency and accuracy of various in-field farm work, including soil sampling [4], crop growth status investigation [1], crop yield identification [2], etc..

Agronomists and farmers usually need to pre-define targets before conducting in-person field trips to save time and effort. Targeting underperforming areas, post-data analyzing, and final decision-making are typically conducted indoors on desktops as they require detailed visualization and complex data manipulation. However, to the best of our knowledge, no prior research has attempted communicating AR visualization tools and other traditional visualization tools to support a complete PF decision-making process.

Therefore, we developed the first PF observation creation and visualization system that communicates an outdoor mobile AR client

with an indoor web client to support a complete underperforming area investigation process seamlessly. The outdoor mobile AR client provides a simplified and head-mounted AR interface to support efficient underperforming area navigation, observation, and crop treatment. In contrast, the web-based interactive multi-layer mapping tool provides sophisticated and thorough data visualization to help agronomists identify underperforming areas and conduct an in-depth analysis of the observation data collected from the field trips for final decision-making. This paper mainly focuses on the design of the outdoor mobile AR client.

2 SYSTEM DESCRIPTION

The outdoor mobile AR client is developed on a Microsoft HoloLens to allow hands-free interactions during the field trip. The AR client consumes the underperforming area data collected from the web client and supports in-situ observation creation and field-data visualization for these areas. By providing a simplified and intuitive in-field navigation and observation data creation interface, this system allows farmers and agronomists to efficiently investigate the issues that caused the underperformance of these targeted areas and accurately record the observed issues.

2.1 In-Field AR Navigation to Underperforming Areas

The web client user allows the agronomist to identify and mark specific underperforming areas (Fig. 1 (b)) by conducting a higher-level multi-layer analysis of the crop yield frequency and soil nutrient clustering maps [3]. An underperforming area may show significantly lower crop yield or nutrient deficiencies. The web client generates a zone and exports it as a GeoJSON/SHP file of the identified underperforming areas, which is then sent to the mobile AR client to support the targeted field trips for the agronomists or farmers.

The AR mobile client tracks the user's real-time GPS location by scanning the Bluetooth Low Energy (BLE) advertisement sent by an Android mobile phone [4]. A customized shade may be used to overcome the brightness issue for outdoor HoloLens usage. It visualizes the field minimap (Fig. 1 (a)) for the user to select a neighborhood field to start a field trip. Then, the AR client visualizes the yield frequency map of the selected field and navigates the user to the underperforming areas by in-situ overlaying the navigation routine to the underperforming area on the field (Fig. 1 (c)). To avoid the user repeating investigating the same area, the mobile AR client real-time records the routine that the user has passed by with the overlaid footprints on the ground (Fig. 1 (d)).

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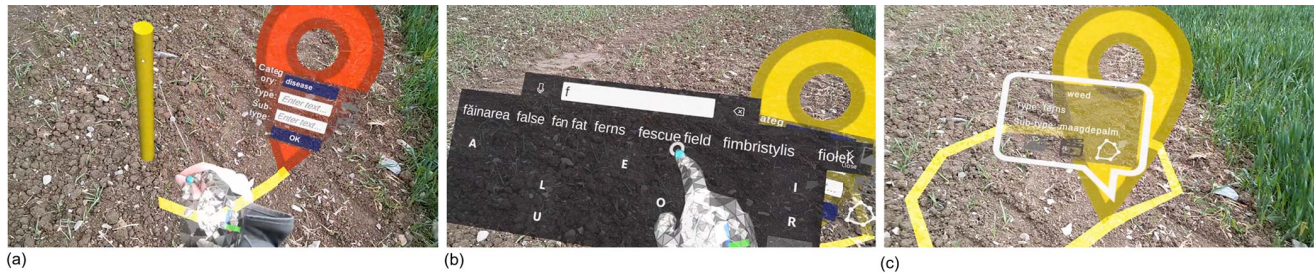


Figure 2: (a) Create AR Annotations to Record Observations for Problematic Areas; (b) Customized AR Keyboard to Simplify the Long-Complex Term Input; (c) In-Situ Visualizing Historical Observations

2.2 In-situ Observation Creation

After arriving at the targeted underperforming area, the user can create AR annotations at identified problematic areas to record the issues they observed there, which we call the *observations* in this paper. By pointing at a problematic area and speaking a voice command, an AR annotation will be created in-situ (Fig. 2 (a)) to allow for observation data input, including problematic area highlighting, observation type inputting, image capturing, video recording, and neighborhood sensor data recording.

2.2.1 Problematic Area Highlighting

To accurately record where the issue is found, this system allows the user to highlight a problematic area for an *observation* using an AR brush. By dragging the AR brush, the user can draw a polygon on the ground to highlight where they observed the issues (Fig. 2 (a)).

2.2.2 Simplified AR Keyboard for Customized-Complex Input

For each *observation*, the system requires the user to input the observation category, type, and sub-type to help the agronomists accurately diagnose the issue. The user can easily select the observation category from a drop-down list, which only includes four categories. However, the option lists of *observation* types and sub-types both contain thousands of long-complex terminologies about crop disease, pests, and weed. Therefore, common input strategies may not work efficiently on an AR headset in an outdoor environment. Accordingly, we modified the original HoloLens MRTK keyboard¹ to simplify the long-complex terminology input by pre-defining the customized terminology lists and filtering the matching terms and following letters on the keyboard (Fig. 2 (b)).

2.2.3 Peripheral Sensor Data Recording

To help the agronomists accurately diagnose the issue according to the environmental parameters, the system automatically records the peripheral relevant sensor data for each *observation*, which includes soil temperature and moisture sensors, sunlight sensor, PH meters, etc.. The system collects peripheral sensor signals using BLE advertising scanning at this stage. It records a snapshot for scanned sensor data when the user creates an *observation*. Accordingly, at the post-analyzing stage, the agronomists may quickly gauge relevant environmental parameters when an observation was created.

2.3 Observation Storage and Reproduction

The system records each AR *observation* as a GeoJSON format, which includes the properties about *observation* category, type, sub-type, the *observation* GPS, creation time, the polygon coordinate list that highlights the problematic area, peripheral sensor data snapshots, and the Uniform Resource Identifiers (URI) of the uploaded

¹<https://docs.microsoft.com/en-us/windows/mixed-reality/mrtk-unity/features/ux-building-blocks/system-keyboard?view=mrtkunity-2021-05>

observation image and video. For each underperforming area, the system records a JSON file that contains a list of GeoJSON geometries of all the *observations* which the system uploads to a cloud service using Microservices.

Using these JSON files, the web client visualizes the *observations* of corresponding underperforming areas on the yield frequency or nutrient clustering maps to allow for in-depth data analyzing and issue diagnosing. Accordingly, agronomists may make crop treatment decisions to address the diagnosed issues. To support future re-visiting to the underperforming areas for crop treatments, the mobile AR client allows for in-field observation reproduction by in-situ visualizing historical observations (Fig. 2 (c)).

3 CONCLUSIONS

This system combines a web-based interactive multi-layer mapping tool and a mobile AR in-field data in-situ visualization tool to support a complete precision farming decision-making process. The agronomists can conduct complex field data analysis to identify the underperforming areas with the web client, and investigate the targeted underperforming areas with the mobile AR in-field navigation and observation features. Next, the agronomists may analysis the observation data on the web clients to make decisions on crop treatment and fertilization. Finally, they may revisit the underperforming area to conduct crop treatments using the system.

By exploiting the unique advantages of both visualization strategies, the system allows the agronomists to complete complex and time-consuming data analysis tasks indoors with the sophisticated web client, while conducting outdoor investigations efficiently with the simplified and intuitive mobile AR client.

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